

Month 1 Project – Full Stack Development Fundamentals

Project Title: Personal Portfolio & Task Management Web Application

Objective

The goal of this project is to help students apply the concepts learned during Month 1 of the Full Stack Development program. Students should build a simple full stack web application that includes front-end development, back-end integration, database connectivity, API handling, authentication basics, and deployment preparation.

This project is designed for beginners who have completed the first four weeks of training and are now ready to combine all concepts into one working application.

Project Overview

Students must develop a **Personal Portfolio & Task Management Web Application**.

The application should allow users to:

- View a personal portfolio/profile section
- Add, edit, delete, and manage tasks
- Store task data using a database
- Connect front-end with back-end APIs
- Use responsive UI design
- Maintain project structure using Git & GitHub

This project should demonstrate understanding of both front-end and back-end fundamentals learned during Month 1.

Technologies to be Used

Front-End

Students should use:

- HTML5
- CSS3
- JavaScript (ES6+)
- React.js

UI Requirements

The application must include:

- Responsive design
- Navigation bar
- Cards or sections
- Forms
- Buttons and interactions
- Mobile-friendly layout
- Clean and professional UI

Students are encouraged to use:

- Flexbox
- CSS Grid
- Media Queries

Back-End Requirements

Students should use:

- Node.js
- Express.js

The server must:

- Handle API requests
- Perform CRUD operations
- Connect with database
- Manage routes properly
- Return JSON responses

Database Requirements

Students can use either:

- MongoDB
or
- MySQL / PostgreSQL

The database should store:

- User information
- Task details
- Task status
- Created date/time

Core Features to Implement

1. Portfolio Section

Include:

- Student name
- About section
- Skills section
- Projects section
- Contact information

2. Authentication Basics

Students should implement simple authentication concepts such as:

- Login page
- Registration page
- Protected routes (basic level)

JWT authentication can be used if possible.

3. Task Management Module

Users should be able to:

- Add tasks
- Edit tasks
- Delete tasks
- Mark tasks as completed
- View all tasks

4. API Integration

Students must create REST APIs for:

- Creating tasks
- Reading tasks
- Updating tasks
- Deleting tasks

Proper API testing should be done using Postman or similar tools.

5. Version Control

Students must:

- Create a GitHub repository
- Push code regularly
- Use proper commit messages
- Maintain clean project structure

Folder Structure Suggestion

Students should maintain separate folders for:

- Front-End
- Back-End
- Database configuration
- Routes
- Controllers
- Components
- Assets

The project structure should be organized and readable.

UI Expectations

The application should:

- Look modern and clean
- Work properly on mobile and desktop
- Have smooth navigation
- Avoid unnecessary complexity
- Maintain proper alignment and spacing

Students are encouraged to focus more on functionality and clean structure rather than advanced animations.

Deployment Preparation

Students should prepare the project for deployment by:

- Fixing errors and warnings
- Removing unused files
- Testing APIs properly
- Checking responsive design
- Verifying database connection

Optional deployment platforms:

- Vercel
- Netlify
- Render

Submission Requirements

Students must submit:

- GitHub Repository Link
- Project Screenshots
- Project Explanation Document